ASEAN GEOAI FUSION 2026

AGAIF 2026 BOOTCAMP 1: FOUNDATIONAL GEOAI

Introduction to GeoAI & Geospatial Session 1

Online
7 Juy 2026 (Tuesday)

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WHAT DOES GIS STANDS FOR?

GIS = GEOGRAPHIC INFORMATION SYSTEM

GEOGRAPHIC
Study of earth and where things are. *(Geography)*



INFORMATION
For capturing, managing, analyzing and understanding information.



SYSTEM
A set of software, hardware and data.





Attribute of Feature

ID	Location	Type	Population	Area (km²)	Value
001	City Center	Urban	125,000	45.2	85.6
002	River	Water	–	12.8	62.3
003	Forest	Vegetation	–	230.5	76.1
004	Residential	Urban	35,600	18.3	68.4
005	Agricultural	Farm	2,150	150.6	52.7
...	...	...	...	...	...

GIS WORKS BY INTEGRATING DATA, TECHNOLOGY & ANALYSIS

INPUT



Spatial and attribute data from various sources

GIS



Stores, manages and organizes data

ANALYSIS



Analyzes and models data to discover insights

OUTPUT



Generates information visualized as maps

GIS HELPS US TO:

- Understand our world
- Make better decisions
- Solve real-world problems
- Plan for a sustainable future

GIS connects where things are with the information we need to make a difference.

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INTRODUCTION

Geographical Information System

- The acronym GIS stands for ***Geographical Information System.***
- GIS consists of two distinct disciplines, namely, **Geography** and **Information System**.



GIS

GEOGRAPHY

The study of the Earth, its features, locations, and spatial relationships.

&

INFORMATION SYSTEM

The collection, storage, management, analysis, and delivery of information.

GIS integrates Geography and Information System to capture, manage, analyze and visualize spatial data for **better decisions and a smarter world.**

WHAT IS GEOGRAPHY?

The scientific study of the **geospatial pattern** and **process**.



Understanding **where** things are,
why they are there, and **how** they change.

GLOBAL PERSPECTIVE
Geography helps us explore the diverse places and patterns across our world.

PATTERNS & PROCESSES
It examines spatial patterns and the natural and human processes that shape them.

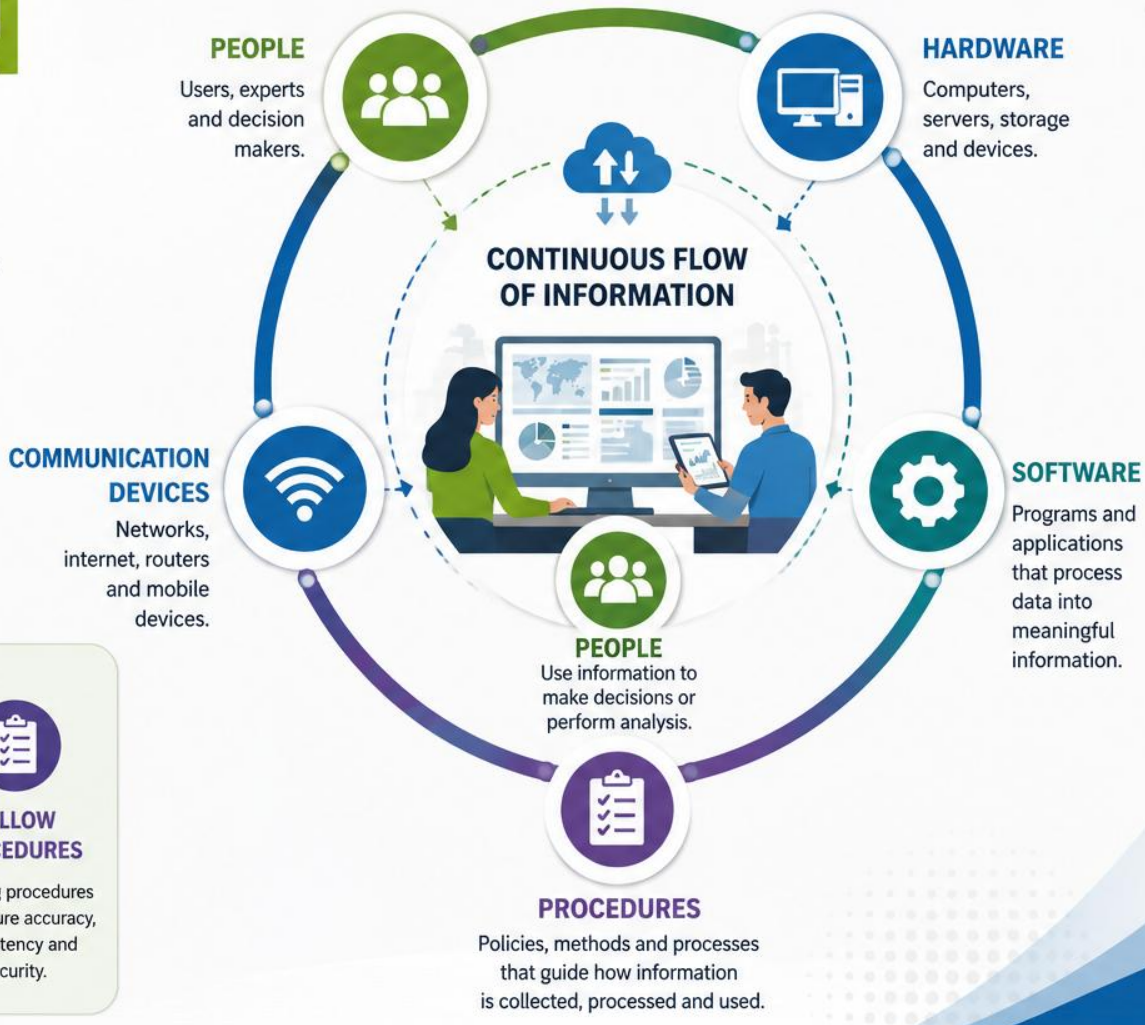
PEOPLE & PLACES
It connects people, cultures, environments and economies across space.

BETTER DECISIONS
Geography provides insights for solving real-world challenges and planning a sustainable future.

Geography helps us make sense of our world and create a better future for all.

WHAT IS INFORMATION SYSTEM?

- It is an interactive combination of **people**, computer **hardware** and **software**, **communication devices** and **procedures** designed to provide continuous flow of information to the people who need information to make decisions or perform analysis.



THE POWER OF INFORMATION SYSTEMS

CONNECT PEOPLE
Bringing people together to share and use information.

ENABLE TECHNOLOGY
Integrating hardware and software to process data efficiently.

COMMUNICATE SEAMLESSLY
Ensuring reliable communication for real-time information exchange.

FOLLOW PROCEDURES
Applying procedures that ensure accuracy, consistency and security.

WHAT IS A GIS?

What is GIS?

- A geographical information system is a system which **integrates** hardware, software and data for **capturing**, **managing**, **analyzing** and **visualizing** all forms of geographically referenced information.

- In simplest terms it is the merging of **cartography**, **statistical analysis** and **database technology**.





CAPTURE
Collect spatial data from the real world.

MANAGE
Organize and store data efficiently in databases.

ANALYZE
Discover patterns, relationships and trends.

VISUALIZE
Create maps and visuals to communicate and make decisions.

GIS turns data into insight and insight into better decisions.

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HISTORY & DEVELOPMENT OF GIS

From early spatial thinking to a smarter world.

1832

The Beginning

The development of GIS was first started in 1832.

One of the first applications of spatial analysis is in Epidemiology.

1854

John Snow & Cholera Outbreak

In 1854 John Snow depicted a cholera outbreak in London using points to represent the locations of some individual cases, possibly the earliest use of a geographic methodology in epidemiology.

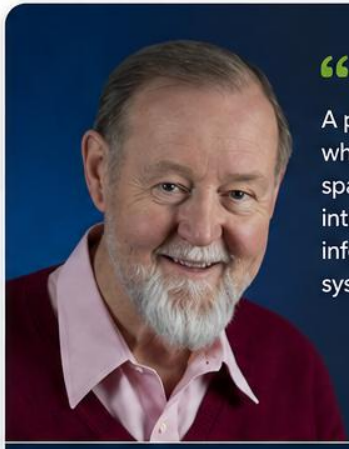


1960

The Modern Era

The year 1960 saw the development of the world's first true operational GIS in Ottawa, Ontario, Canada by the federal dept. of forestry and rural development, developed by **Dr. Roger Tomlinson (Father of GIS),** it was called the Canada Geographic Information System (CGIS).





> "A pioneer who turned spatial thinking into a powerful information system."

Dr. Roger Tomlinson
Father of GIS

Developer of the world's first operational GIS (1960)
Canada Geographic Information System (CGIS)

GIS has evolved from early spatial analysis to the foundation of modern decision-making.

- Understand the past
- Analyze the present
- Visualize the patterns
- Shape the future

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TRADITIONAL PRESENTATION

- **Spatial information** has always been a part of our lives, its just that we **never emphasis it**
- **Map** is an effective way of describing and recording **spatial relationships**



GEOGRAPHIC INFORMATION

NARROW SENSE

Geography can be divided into “geo” and “graphy”, where “geo” is anything that is on the Earth and “graphy” is the recording of those things. Added together is **describing and recording things on the Earth.**

BROAD SENSE

All visible characteristics above or below the surface of the Earth, and all phenomenon or incidents that describe **spatial characteristics.**



FROM MAPS TO MEANING, GEOGRAPHY HELPS US UNDERSTAND OUR WORLD AND SHAPE A BETTER FUTURE.

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FUNDAMENTALS OF GEOGRAPHIC INFORMATION

Geographic information describes our world through **four essential fundamentals.**

01

GEOGRAPHICAL LOCATION

- **Where:** the location of the geographic information. Geographic information is a form of spatial information, thus the spatial location is required to be defined.

02

ATTRIBUTES

- **What:** describes the geographic information. Geographical information also includes non-spatial information which is attributes.

03

SPATIAL RELATIONSHIP

- **How:** the relationship between objects in space.

04

TIME

- **When:** the relationship between objects of different time periods. This is more difficult to deal with. Example: detecting land use changes and possible illegal developments in natural reserves – **Example**



Understanding these fundamentals helps us capture, analyze and manage our world better for a smarter tomorrow.

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EXAMPLES OF GEOGRAPHIC INFORMATION

Geographic information has two main components:
where things are (spatial data) and what they are like (attributes).

SPATIAL DATA

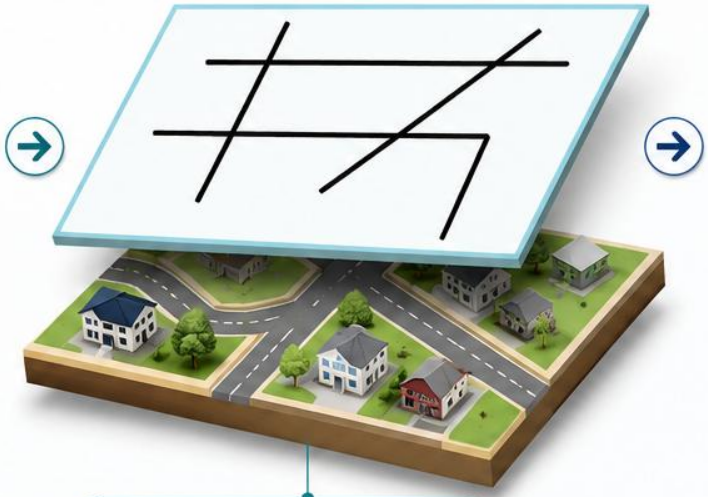
Shows where features are located on the Earth.



Example: Coastal road network showing locations of roads and towns

ROAD MAP

A map is an effective way of describing and recording spatial relationships.



Example: Road map showing connections and intersections

ATTRIBUTE

Provides details or characteristics about the features.

Table - towns

FID	VILLAGE	DAERAH	DISTRICT	CITY
0	Cheencg	Daaa	Daaa	Tapiei City
1	Chetappa	Daaa	Daaa	Tapiei City
2	Deee	Daaa	Daaa	Tapiei City
3	Deea	Daaa	Daaa	Tapiei City
4	Deena	Daaa	Daaa	Tapiei City
5	Pertubang	Daaa	Daaa	Tapiei City
6	Paube	Daaa	Daaa	Tapiei City
7	Pinbtu	Daaa	Daaa	Tapiei City
9	Temamping	Daaa	Daaa	Tapiei City
10	Oangpaka	Daaa	Daaa	Tapiei City
11	Osangpaka	Daaa	Daaa	Tapiei City
12	Orangpa	Daaa	Daaa	Tapiei City
13	Gubang	Daaa	Daaa	Tapiei City
14	Gubanso	Daaa	Daaa	Tapiei City

Show: All | In 1443 record(s), 0 record(s) selected | Options

Example: Attribute table showing information about each town

By combining spatial data, maps, and attributes, we can better understand, analyze and manage our world.

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PREDICAMENTS OF TRADITIONAL METHODS

Outdated practices. Critical gaps. Limited impact.

a) DATA COLLECTION
Data recorded at different times can be **inconsistent**

b) DATA MANAGEMENT
Maps are easily **worn out or lost** over time

c) DATA ACCESS
Possessed by different institutes, therefore is **difficult to access**

d) DATA ANALYSIS
Often **lacks** the required **spatial** information for analysis

e) DATA DISPLAY
Decision-makers **cannot fully comprehend** the relationship between information and space



TRADITIONAL METHODS LIMIT INSIGHT.
MODERN SOLUTIONS UNLOCK IMPACT.

It's time to move beyond limitations and embrace **smarter, spatial solutions.**

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Key Player and GIS Software



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THE EVOLUTION OF GIS → SPATIAL INTELLIGENCE INFRASTRUCTURE

2010–2030

2010
Digital Mapping



Dominant Workflow:
- static mapping
- desktop GIS
- layer visualization

Primary Users:
- GIS specialists
- planners
- cartographers

Technology Shift:
- digitization of spatial data
- transition from paper maps

2020
Enterprise GIS Platforms



Dominant Workflow:
- web GIS
- dashboards
- enterprise integration

Primary Users:
- government departments
- utilities
- enterprise operations teams

Technology Shift:
- cloud GIS
- mobile GIS
- system interoperability

2025
GeoAI + Real-Time Systems



Dominant Workflow:
- AI-assisted analysis
- real-time monitoring
- predictive workflows

Primary Users:
- operations centers
- smart city teams
- analysts
- infrastructure managers

Technology Shift:
- IoT integration
- AI copilots
- streaming geospatial data

2030
Spatial Intelligence Infrastructure



Dominant Workflow:
- autonomous spatial reasoning
- conversational analytics
- decision orchestration

Primary Users:
- entire enterprises
- autonomous systems
- AI agents
- public infrastructure networks

Technology Shift:
- invisible GIS infrastructure
- spatial AI ecosystems
- continuous intelligence systems

MAPS → PLATFORMS → INTELLIGENCE → INFRASTRUCTURE

FROM MAPS TO INTELLIGENCE. FROM DATA TO IMPACT.

GIS is evolving into the foundation of a smarter, more connected, and autonomous future.

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How GIS Turns Climate Data Into Risk Maps

From raw data to **actionable insights** for a safer, resilient future.

1

HAZARD LAYERS IN GIS

- Rainfall
- Temperature
- Flood Extent
- Soil Moisture
- NDVI



Collect and prepare climate and environmental data to understand potential hazards.

2

EXPOSURE LAYERS IN GIS

- Population
- Cities
- Roads
- Crops
- Land Use

Map the elements that are **exposed** to potential hazards.

3

RISK = HAZARD × EXPOSURE

The **foundation** of climate risk assessment.

4

GIS SPATIAL OVERLAY

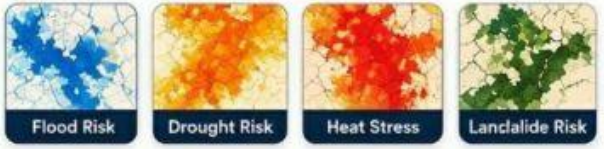
- ArcGIS
- QGIS
- Raster + Vector Analysis
- Pixel-by-Pixel Calculation

Overlay and analyze data using **advanced** spatial techniques.

5

CLIMATE RISK MAPS

- Flood Risk
- Drought Risk
- Heat Stress
- Landslide Risk
- Vulnerability Zones



GIS integrates climate hazards and exposure to turn data into **actionable risk insights.**

Better Decisions | Stronger Communities | Sustainable Future

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Learn GIS

HOW GIS HELPS CHOOSE THE BEST LOCATION.

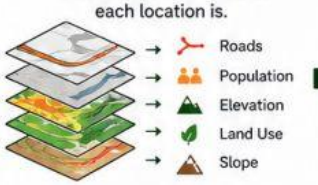
GIS turns data into decisions to find the most suitable place for what matters most.

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THE 3 KEY STEPS

1 SUITABILITY ANALYSIS

Combines multiple factors to determine how suitable each location is.



Evaluate many factors at once, not just one.

2 RANKING CRITERIA

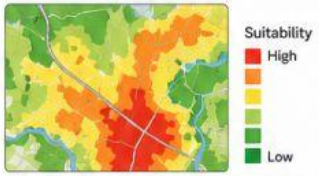
Rank each factor by importance to your goal.

CRITERIA	IMPORTANCE (WEIGHT)
Proximity to Roads	25%
Population Density	30%
Elevation	15%
Land Use Compatibility	20%
Slope	10%

Different goals = different weights.

3 WEIGHTED SCORES

Combine and calculate a score for every location.



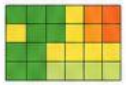
Higher score = more suitable location.

EXAMPLE: FIND THE BEST PLACE FOR A NEW HOSPITAL

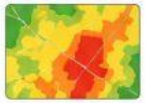
1. INPUT DATA
Collect important layers.



2. STANDARDIZE
Convert all data to a common scale.



3. WEIGHT & SCORE
Apply weights and calculate scores.



4. SUITABILITY MAP
See the best locations.



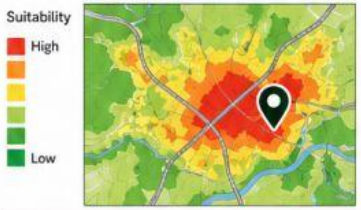
5. BEST LOCATION
Choose the top suitable site.



EXAMPLE CRITERIA & WEIGHTS

CRITERIA	DESCRIPTION	WEIGHT
Proximity to Roads	Closer to major roads is better	25%
Population Density	Closer to more people is better	30%
Elevation	Moderate elevation is preferred	15%
Land Use Compatibility	Compatible land use is better	20%
Slope	Gentle slopes are better	10%
TOTAL		100%

REAL-WORLD EXAMPLE



The red areas are the most suitable locations.

Decision-makers can choose the best site with confidence.

WHY IT MATTERS

- Data-driven decisions
- Saves time & resources
- Reduces risk & uncertainty
- Better services for communities
- Stronger outcomes, smarter investments

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GIS PROCESS

TURNING DATA INTO DECISIONS THROUGH SPATIAL INTELLIGENCE

1 DATA INPUT



- Maps
- Surveys
- GPS / Field Data
- Remote Sensing
- Tabular Data

2 PROCESSING



- Data Cleaning
- Integration
- Transformation
- Georeferencing
- Quality Check

3 ANALYSIS



- Spatial Queries
- Overlay Analysis
- Modeling
- Patterns & Trends
- Decision Support

4 DISPLAY



- Maps
- Charts
- Dashboards
- Reports
- Web Visualization



BETTER DATA BETTER INSIGHT | **SMARTER ANALYSIS STRONGER DECISIONS** | **LOCATION INTELLIGENCE REAL WORLD IMPACT**

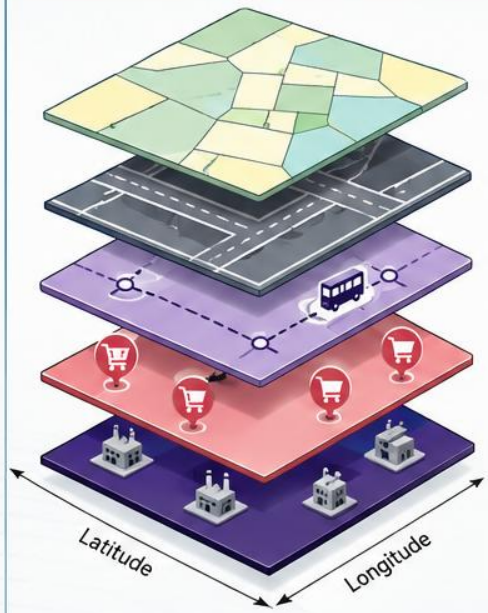
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LAYERS IN GIS (*Vector and Raster*)

GIS ORGANIZES THE REAL WORLD INTO MEANINGFUL LAYERS FOR BETTER UNDERSTANDING AND DECISION-MAKING

REAL WORLD, ORGANIZED IN LAYERS



CENSUS TRACTS
Boundaries of population areas

ROADS
Street and road networks

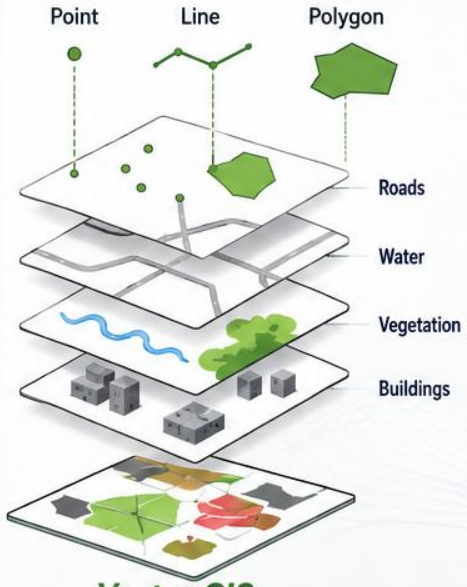
BUS ROUTES
Public transportation paths and stops

SHOPPING CENTERS
Retail and commercial locations

INDUSTRIAL SITES
Factories and industrial facilities

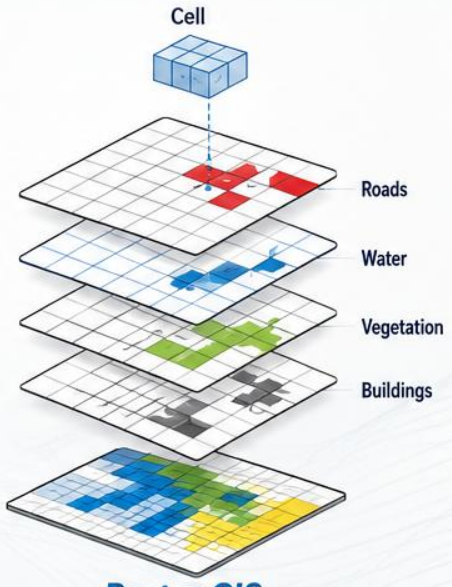
TWO WAYS TO REPRESENT THE WORLD IN GIS

VECTOR GIS



Vector GIS

RASTER GIS



Raster GIS

LAYERS SIMPLIFY complex data by organizing it.

VECTOR CAPTURES exact locations, shapes and connections.

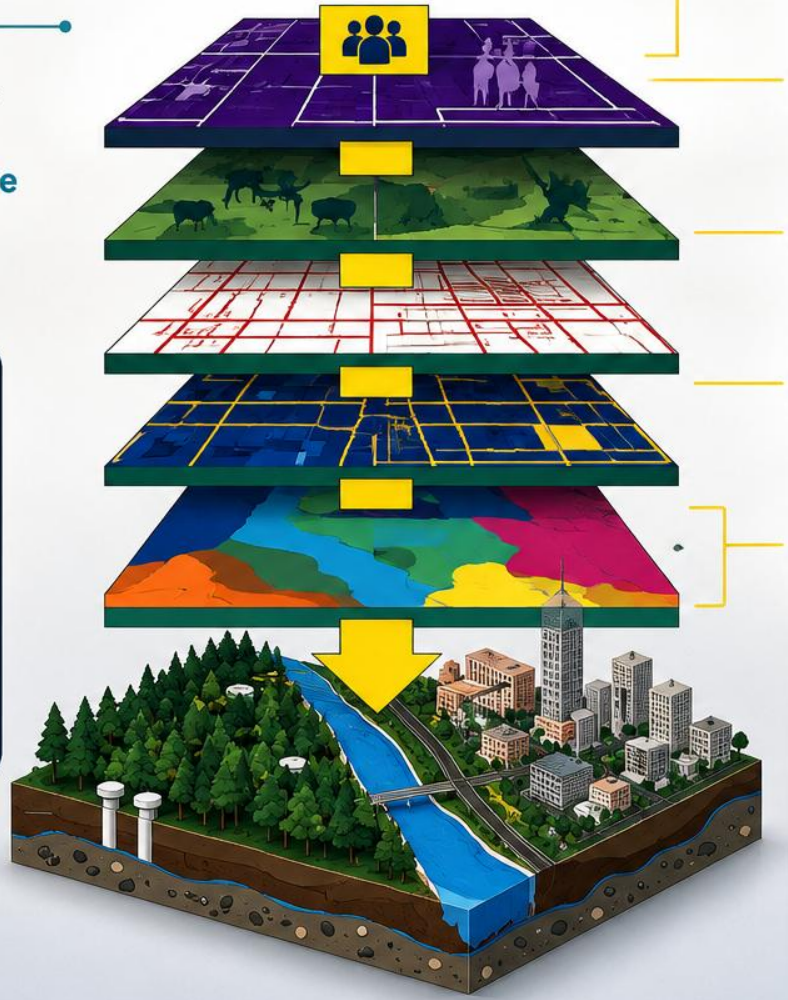
RASTER REPRESENTS continuous surfaces in cells.

TOGETHER, they provide a complete picture.

BETTER INSIGHT. BETTER PLANNING. BETTER FUTURE.

COMBINATION OF DATA LAYERS

Powerful decisions come from **integrating multiple perspectives.**



SOCIAL FACTORS
Population, demographics, community assets, and human activities.

BIODIVERSITY
Habitats, species distribution, ecosystems, and natural resources.

ENGINEERING
Infrastructure plans, utilities, design constraints, and technical layouts.

LAND USE
Zoning, development patterns, existing land use and planning.

ENVIRONMENTAL CONSIDERATIONS
Soil, slope, water resources, climate, and environmental hazards.

STRONGER TOGETHER.
BETTER OUTCOMES.

Combining data layers reveals insights that individual layers alone cannot show.

INTEGRATE. ANALYZE. DECIDE.
Combining these layers in GIS provides a holistic view for smarter planning, sustainable development, and better decision-making.

BETTER PLANNING | SUSTAINABLE DEVELOPMENT | DEEPER INSIGHTS | STRONGER DECISIONS

MODEL DATA: RASTER vs VECTOR

Two ways to model the real world in GIS

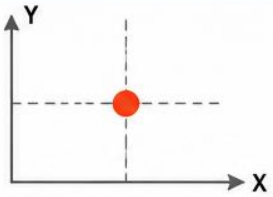
VEKTOR (Vector Data)

Represents features using coordinates

Point

POINT

A single location defined by coordinates (X, Y)



Line

LINE

A series of connected points forming a path



Polygon

POLYGON

A closed shape defined by its boundary (vertices)



VS

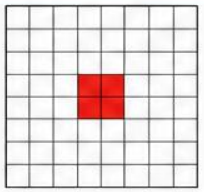
RASTER (Raster Data)

Represents features using a grid of cells (pixels)

Point

POINT

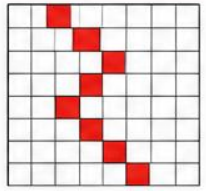
A cell (pixel) represents a location



Line

LINE

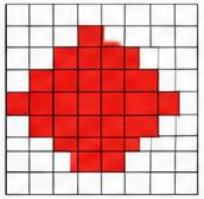
A sequence of cells (pixels) represents a path



Polygon

POLYGON

A group of connected cells (pixels) represents an area



VECTOR DATA

- Precise boundaries
- Smaller file size (for discrete data)
- Best for objects and networks

DIFFERENT REPRESENTATION, SAME PURPOSE:
Model the real world for better decisions.

RASTER DATA

- Simple data structure
- Better for continuous surfaces
- Ideal for imagery and analysis

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COORDINATE SYSTEM vs PROJECTION SYSTEM

Understanding the difference between locating positions on the Earth and mapping them on a flat surface

Naratif Baharu UKM — Universiti Watan Kita

A coordinate system tells us where a location is on the Earth.
A projection system tells us how the curved Earth is transformed onto a flat map.

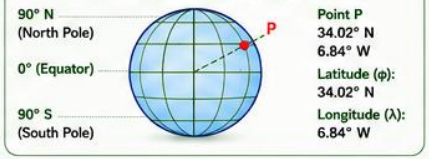
COORDINATE SYSTEM

Usually Geographic Coordinate System (GCS)



- **What it uses** — Latitude & Longitude
- **Surface** — Curved Earth / ellipsoid
- **Units** — Degrees (°)
- **Coordinate example** — 34.02° N, 6.84° W
- **Best for** — Global location, GPS, data storage
- **Common example** — WGS 84

Parallels & Meridians on the Globe



✓ Shows location directly on the Earth.

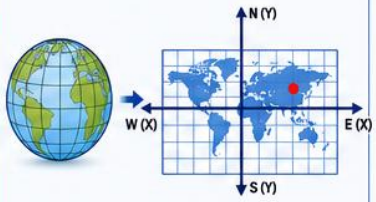
PROJECTION SYSTEM

Usually Projected Coordinate System (PCS)



- **What it uses** — X & Y coordinates
- **Surface** — Flat map
- **Units** — Meters or feet
- **Coordinate example** — 412000 m E, 3765000 m N
- **Best for** — Mapping, analysis, distance & area
- **Common examples** — UTM, Lambert, Mercator

From Curved Earth to Flat Map



✓ Transforms the Earth for practical mapping.

COORDINATE SYSTEM	COMPARE	PROJECTION SYSTEM
Globe (Earth)	Shape	Flat map
Latitude / Longitude	Coordinates	X / Y
Degrees (°)	Units	Meters / Feet
Locate positions	Main use	Measure & analyze
WGS 84	Example	UTM, Lambert, Mercator

Earth → Coordinate Reference → Projected Map



IN SHORT

Coordinate systems define positions on the Earth.
Projection systems convert those positions to a flat map for analysis and cartography.

COMMON COORDINATE & PROJECTION SYSTEMS

Coordinate System (GCS)

- **WGS 84** – Global standard (GPS)
- **NAD 83** – North America
- **GDA 2020** – Australia
- **Kertau RSO / Malay RSO** – Malaysia

Projection System (PCS)

- **UTM** – Universal Transverse Mercator
- Lambert Conformal Conic
- Mercator
- Albers Equal Area

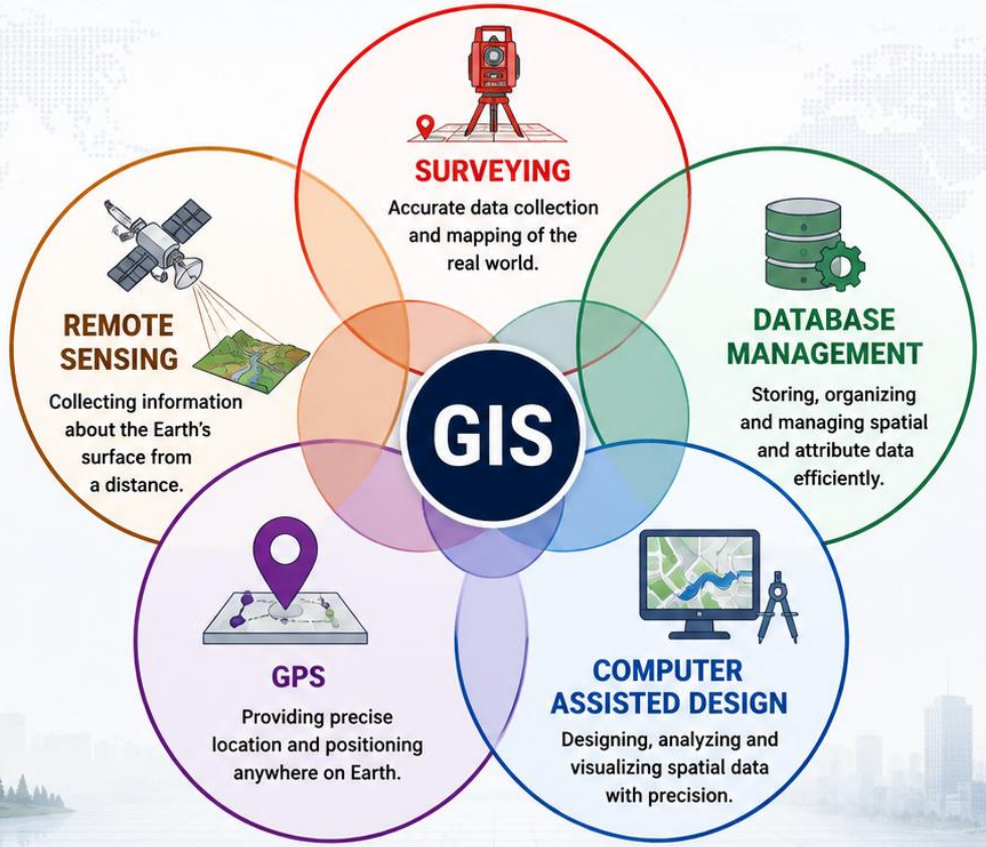
Why it matters?

Using the right system ensures accurate positioning, measurement, analysis and better decision-making in GIS, mapping and spatial planning.

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GIS TECHNOLOGIES

Integrating Multiple Technologies • Empowering Better Decisions



SEE THE WORLD FROM ABOVE

Remote sensing captures up-to-date information for better insights.

KNOW EXACTLY WHERE

GPS delivers the accuracy you need for reliable spatial information.

MANAGE DATA SMARTER

Robust database management ensures data integrity, security and access.

DESIGN WITH CONFIDENCE

CAD tools bring ideas to life with accurate spatial planning and visualization.

FIVE TECHNOLOGIES. ONE POWERFUL SYSTEM.

GIS brings them together to solve real-world challenges.

Better Data Integration | Stronger Analysis | Smarter Decisions | Greater Collaboration | Sustainable Future

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ADVANTAGES OF A GIS

Smarter Data. Better Decisions. Stronger Impact.

01 DATA STORED IN DIGITAL FORMAT
Data stored in digital format so physically more compact.

02 LARGE QUANTITIES OF DATA
Large quantities of data can be retrieved at greater speed.

03 REDUCTION IN DATA REDUNDANCY
Eliminates duplicate data and improves efficiency.

04 LOW COST OF MAINTAINING AND RETRIEVING DATA
Reduces costs associated with data storage and access.

05 DATA INTEGRATION
Integrates GPS, remote sensing, tabular data and more for a complete picture.



GIS
Integrating Data, Enabling Insights, Driving Solutions.

06 PROVISION OF COMPLEX SPATIAL ANALYSIS
Performs advanced spatial analysis to solve complex real-world problems.

07 MAINTAINING DATA CONSISTENCY
Ensures accuracy, reliability and consistency of data.

08 CAPABILITY OF DATA UPDATING
Data can be easily updated to reflect the latest changes.

09 CAPABILITY OF DATA PROCESSING AND MODELING
Processes data and builds models to predict and support decisions.

10 ABLE TO IDENTIFY AND QUERY SPATIAL DATA
Easily identify, search and query spatial data for quick insights.

GIS EMPOWERS YOU TO WORK SMARTER, SAVE TIME AND MAKE BETTER DECISIONS.

Better Data Management | Faster Access | Smarter Analysis | Better Decisions | Stronger Impact

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THE ROLE OF GIS

Turning Data into Decisions, Shaping a Better World

LAND, ECOLOGICAL AND NATURAL RESOURCES DATA...

TOPOLOGY, HYDROLOGY AND HUMANITIES DATA...

GIS

SOFTWARE
Powerful tools for spatial analysis and visualization

+

DATABASE
Organized data for storage, retrieval and management



INTEGRATE • PROCESS • ANALYZE • VISUALIZE
Transforming data into meaningful insights

MANAGEMENT
Oversee resources and ensure efficient operations

EXECUTION
Implement plans and carry out field activities

DECISION
Make informed, data-driven decisions with confidence

RESULT
Better Insights, Better Decisions

ADMINISTRATION
Establish policies, allocate resources and provide direction

PLANNING
Develop strategies and spatial plans for the future

ANALYSIS
Analyze spatial patterns and trends to generate valuable insights

INTEGRATE
Multiple data sources in one platform

MANAGE
Data efficiently and securely

ANALYZE
Discover patterns and relationships

DECIDE
Make smarter, faster decisions

IMPACT
Better outcomes for a better world

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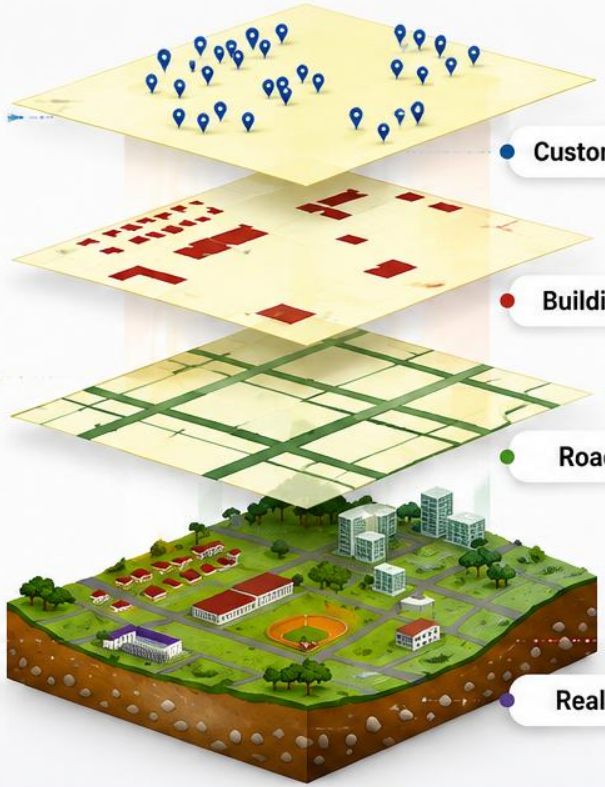
GIS FUNCTIONS

Transforming Data into Decisions

01 DATA COLLECTION
Gather data from various sources about real-world features.

02 DATA STORAGE
Organize and store spatial data in a structured database.

03 DATA QUERY
Search, filter and retrieve the data you need.



04 DATA ANALYSIS
Analyze spatial data to discover patterns, trends and relationships.

05 DISPLAY RESULTS
Visualize data through maps, charts and dashboards.

06 EXPORT RESULTS
Export maps, reports and data for sharing, printing or further use.

From Data to Insight. From Insight to Impact.

Collect Accurate Data → Store Securely → Query Efficiently → Analyze Smartly → Display Clearly → Export Easily

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EXPORT

Present, Share, and Communicate Your GIS Results

Export tools in GIS help you present spatial information clearly and professionally.

1 VISUAL DISPLAY

MAP | TABLE | CHART

2 MAP CONTENT

MAIN BODY
The main map frame that displays your spatial data.

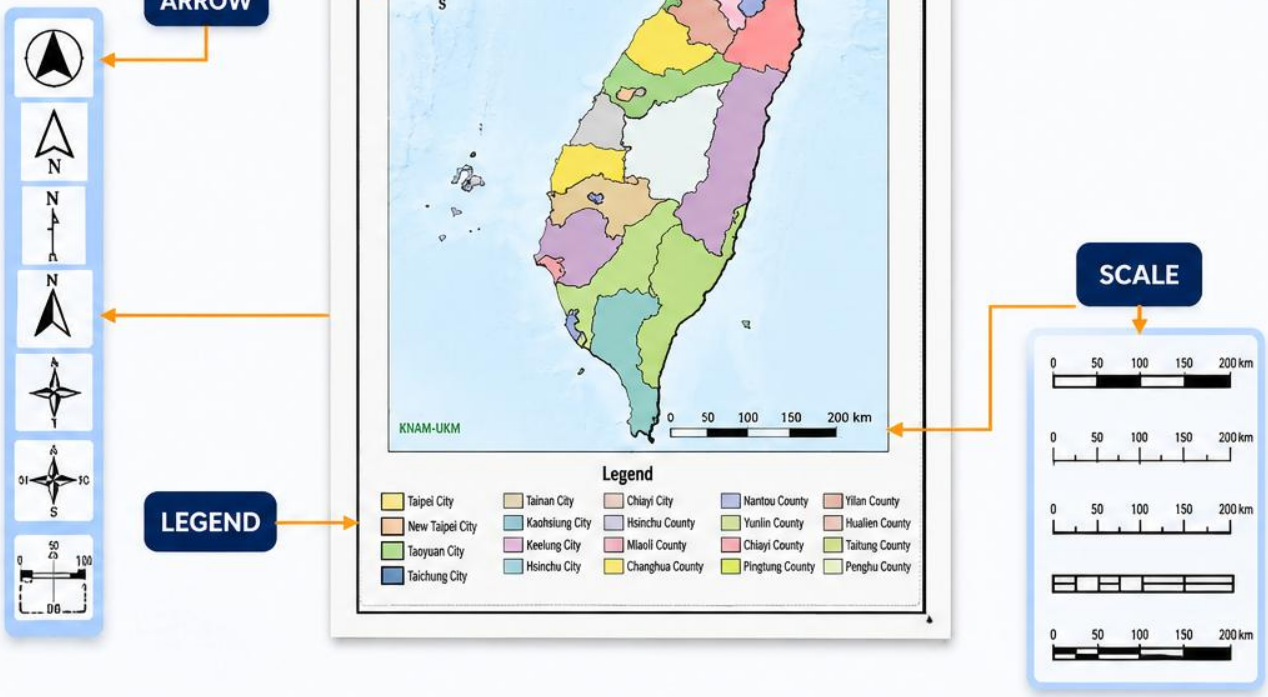
TITLE
Provides a clear title to describe the map.

NORTH ARROW
Indicates the orientation of the map.

SCALE
Shows the relationship between map distances and real-world distances.

LEGEND
Explains the symbols, colors and features on the map.

NEATLINE
A border line that frames the map elements.



EXPORT YOUR GIS RESULTS WITH CLARITY AND IMPACT.

Clear Visualization | Complete Map Content | Effective Communication | Easy Sharing & Reporting

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IMPORTANT PART OF GIS

Powerful Components. Smarter Decisions.

01



ALGORITHM

Sets of rules and procedures that process spatial data and solve complex problems efficiently.

02



SPATIAL ANALYST

Analyzes spatial patterns, relationships and distributions to understand locations and make better decisions.

03



NETWORK ANALYST

Evaluates and optimizes network systems such as roads, utilities and communication for best routes and efficiency.



04



3D ANALYST

Analyzes 3D data and visualizes terrain, buildings and scenes for deeper spatial understanding.

05



STATISTIC ANALYST

Applies statistical methods to spatial data to uncover trends, correlations and significant insights.

06



MODELING

Builds predictive models to simulate scenarios, forecast outcomes and support strategic planning.

TOGETHER, THESE COMPONENTS MAKE GIS
A POWERFUL TOOL FOR SOLVING REAL-WORLD CHALLENGES.

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GIS VS WEBGIS

DESKTOP POWER vs ONLINE ACCESS

WHAT IS GIS?

GIS stands for Geographic Information System. It is used to collect, store, analyze, and visualize spatial or location-based data.

VS

WHAT IS WEBGIS?

WebGIS is GIS on the web. It allows users to access maps, spatial data, and GIS tools through an internet browser.

FEATURE	GIS	WEBGIS
PLATFORM	Desktop software	Web browser
ACCESS	Usually from one computer	Anytime, anywhere
USERS	Mostly technical users	Technical and non-technical users
SHARING	File-based sharing	Online sharing
DATA UPDATE	Manual or local update	Real-time or server-based update
COLLABORATION	Limited	Easier and faster
EXAMPLE TOOLS	QGIS, ArcGIS Pro	Leaflet, Mapbox, GeoServer

SIMPLE DIFFERENCE

GIS is mainly used for advanced spatial analysis.

WebGIS is used to make maps and spatial data accessible online.



WHY WEBGIS MATTERS

WebGIS makes GIS more open, interactive, scalable, and useful for decision-making.

Accessible | Interactive | Scalable | Smarter Decisions

WEBGIS
LEARN. BUILD. SOLVE. LEAD.

CONNECT People | VISUALIZE Places | EMPOWER Decisions

webgis.org

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FUTURE GIS



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IN SHORT, A GIS IS....

01 An information system (hardware, software, data)



02 which enables us to apply lots of analysis models



03 to any geographical datasets



04 for generating derived information



05 that can be visualized as maps.





GIS INTEGRATES DATA, ANALYSIS AND VISUALIZATION TO TURN INFORMATION INTO INSIGHT.

GIS transforms data into meaningful maps to support better decisions for a better world.

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Understanding
where things are.

Describing
what things are like.

Connecting
information to make better decisions.

GIS EMPOWERS US TO
UNDERSTAND TODAY,
ANALYZE DEEPER,
AND PLAN BETTER
FOR TOMORROW.

شكرا *(shukran)*

Tanemmirt
Thank You
Terima Kasih

THE END



INFORMATION HAS VALUE.
GEOGRAPHY GIVES IT MEANING.

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